



UNITED INTERNATIONAL UNIVERSITY

PROJECT REPORT

CareerForge

A Student Career & Opportunity Management Workspace

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1. Introduction

CareerForge is a web-based student career workspace that combines career profile management, CV/document management, job discovery, intelligent job matching, applications, AI-assisted learning paths, learning resources, community participation, student connections, and real-time private chat.

The platform has Student and Admin workspaces. The documented student journey connects registration, profile completion, CV creation, smart job matching, applications, learning, resources, community, connections and chat.

2. Objectives

- Centralize student career preparation and opportunity discovery.
- Maintain structured profiles and reusable CV versions.
- Provide manual search and profile-aware smart job matching.
- Support applications with preserved CV snapshots.
- Provide AI-assisted learning paths and progressive assessments.
- Curate learning resources and skill-based discovery.
- Enable community interaction with automated moderation.
- Enable accepted connections to communicate privately in real time.
- Provide administrative management for jobs, users, resources, events and community content.

3. Project Description

CareerForge treats career preparation as a connected workflow. Profile and CV data feed job matching; applications preserve CV evidence; learning paths and resources support skill development; community and connections support collaboration.

Area	Main Functions
Authentication	Registration, login, role validation, suspended-account blocking
Profile/Career Vault	Profile, CV versions, documents
Jobs/Applications	Job publishing, matching, apply, status tracking
Learning	AI paths, levels, quizzes, progress
Resources/Events	Curated resources and published events
Community	Posts, interactions, moderation
Connections/Chat	Requests, accepted connections, real-time private chat
Admin	Users, jobs, applications, resources, events, moderation

4. System Architecture and Technology Overview

The documented implementation uses a web frontend communicating with backend REST APIs, persistent database tables, a WebSocket channel for chat, and optional external/AI integrations for semantic embeddings, quiz generation, YouTube playlists and job-source synchronization.

Component	Responsibility
Web Interface	Student/admin dashboards and workflows
REST Backend	Validation, business rules, persistence and API responses
Database	Users, profiles, CVs, jobs, applications, learning, resources, community, connections and messages
WebSocket	Live private chat delivery
Gemini	Semantic embeddings and AI learning/quiz support when configured
Groq	Quiz-generation fallback
YouTube API	Optional public playlist search
Job source adapters	Import external job listings and prevent duplicates

The guide documents endpoints for authentication, profiles, jobs, resources, community and a WebSocket chat endpoint.

5. Student-Side Features

5.1 Registration, Login and Profile

Registration uses name, email and password; passwords are BCrypt-hashed. Login validates credentials, role and account status. The profile stores university, degree, graduation year, experience, target role, location, skills, hobbies, bio and photo.

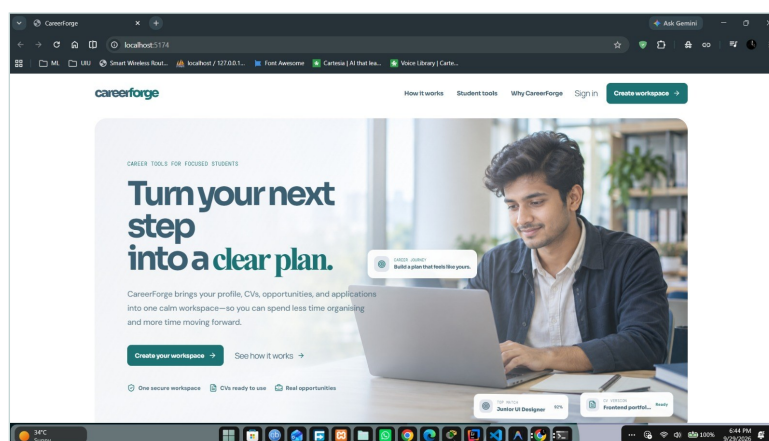


Figure 1. CareerForge landing page.

5.2 Career Vault

Students can create multiple CV versions, select a default, manage documents, and preserve a CV snapshot with each application. The Career Vault lets students build tailored resume versions with a live preview, or import an existing CV from a PDF, DOC or DOCX file; the selected CV is used when applying for a job.

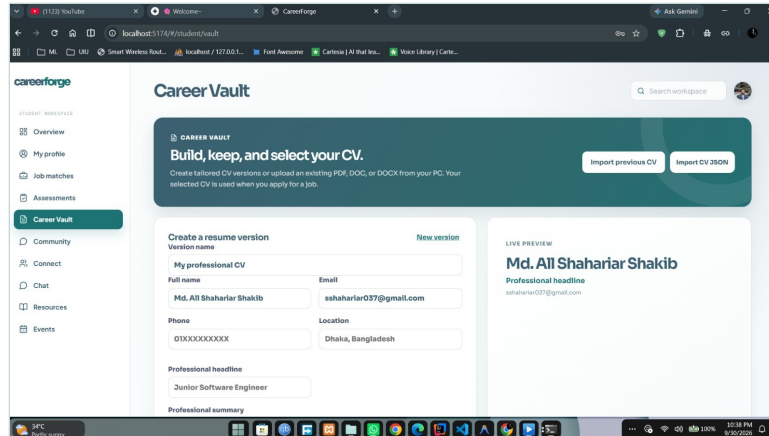


Figure 2. Student Career Vault page with resume version builder and live preview.

5.3 Job Matches

Manual search supports job title/company, location, required skill, work mode and job type. Smart matching compares profile/CV information against published jobs.

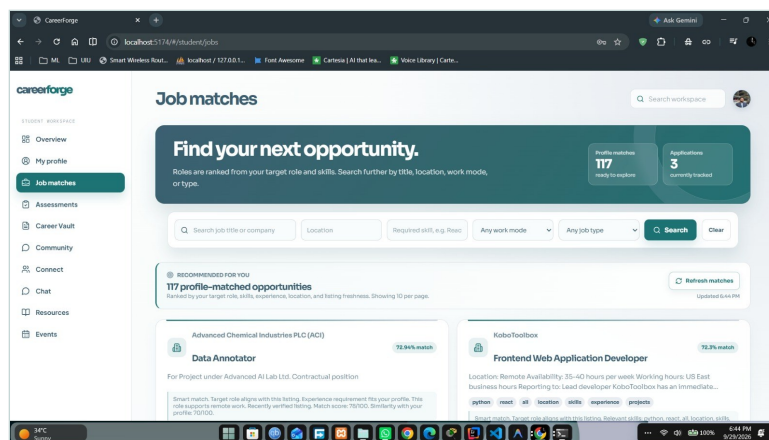


Figure 3. Student Job Matches page.

5.4 Assessments

Learning paths can be created for topics such as Java, React or Data Analyst. Generated quiz attempts are stored, and 70% or above unlocks the next level. Gemini, Groq and local fallback mechanisms are documented.

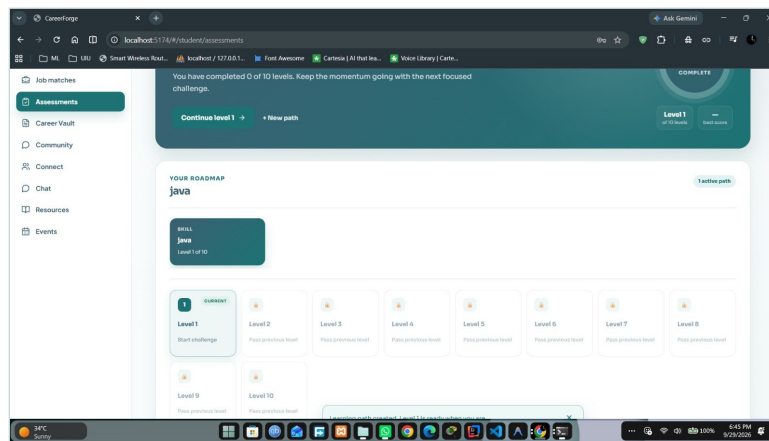


Figure 4. Student Assessments page.

5.5 Resources

Students can search by skill, save resources and mark them complete; progress persists. Optional YouTube search can return public playlists.

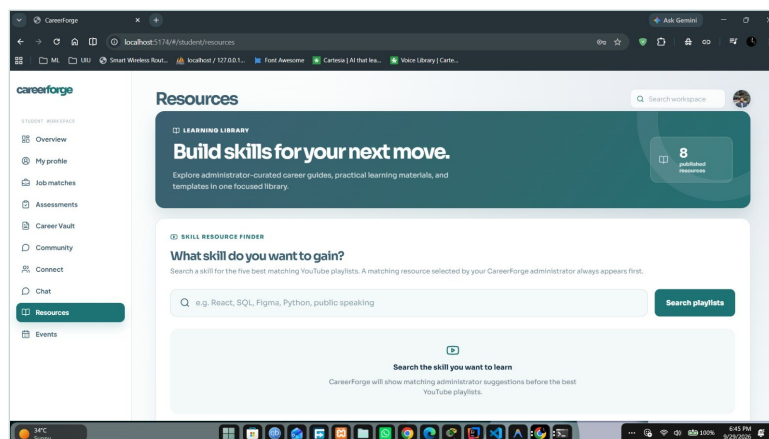


Figure 5. Student Resources page.

6. Admin-Side Features

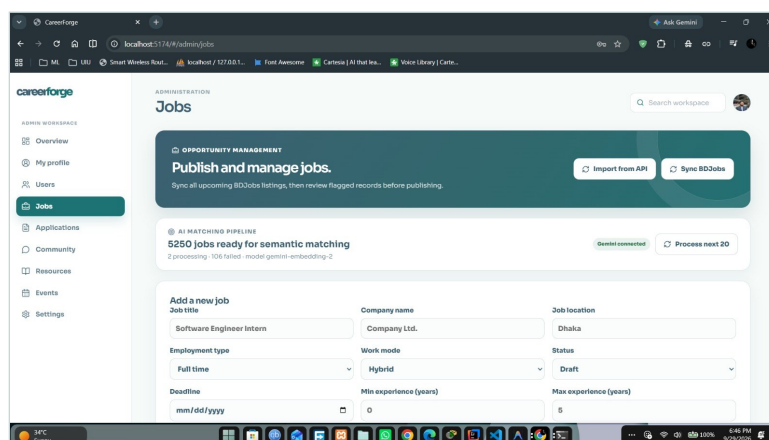


Figure 6. Admin Jobs page.

Admins create and manage jobs, publish listings, synchronize sources, and monitor semantic-processing status.

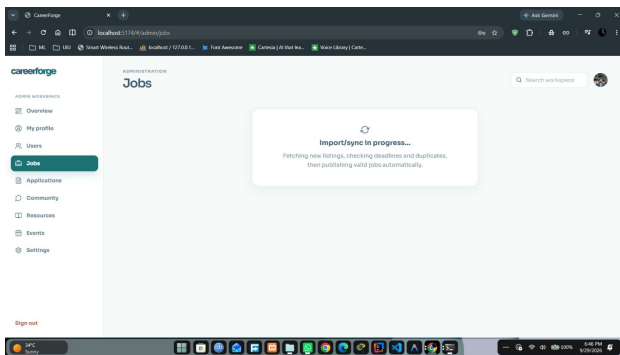


Figure 7. Job import/sync progress.

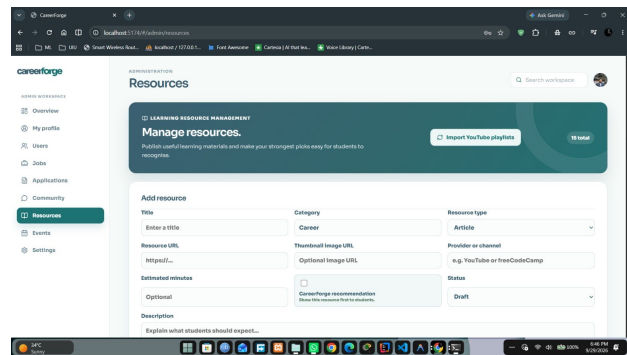


Figure 8. Admin Resource Management.

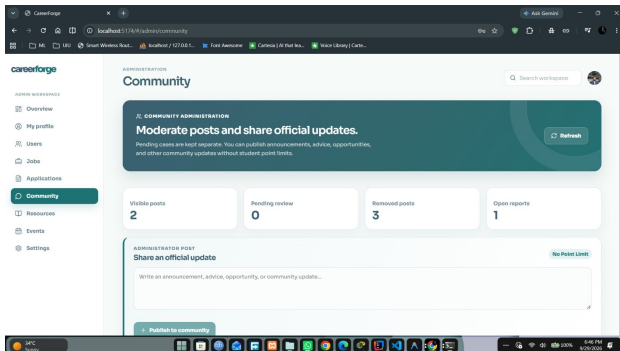


Figure 9. Admin Community moderation dashboard.

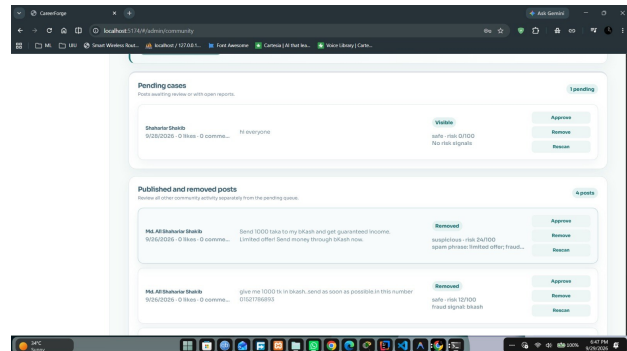


Figure 10. Pending cases and removed posts.

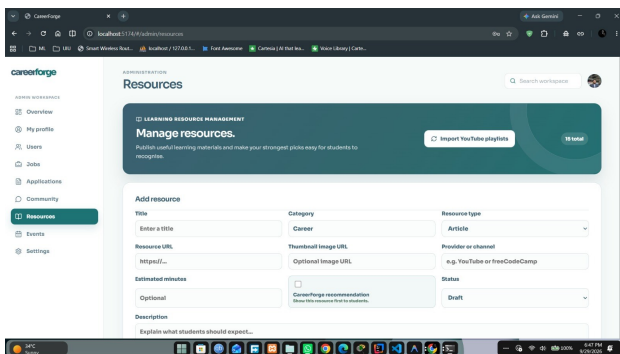


Figure 11. Admin Resources overview.

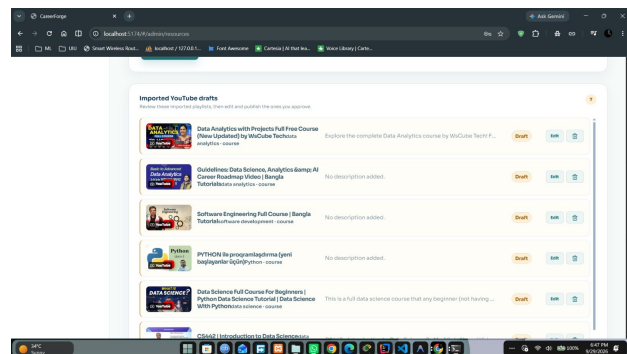


Figure 12. Imported YouTube resource drafts.

7. Smart Job Matching

The documented score uses target role (40), experience (25), skills/CV/interests (20), location/work mode (10), and job freshness (5). When Gemini embeddings are available, final ranking is documented as 40% rule-based score plus 60% semantic similarity; otherwise rule-based matching continues to work.

Factor	Max	What it checks
Target role	40	Role and job-title/description terms
Experience	25	Profile experience vs. job range
Skills/CV/interests	20	Relevant terms
Location/work mode	10	Preference or remote support
Job freshness	5	Recently verified/created job

8. AI Learning Path and Assessment

Students create topic-based paths and unlock levels through assessment. Quiz generation falls back from Gemini to Groq and then to a local fallback if external services fail.

9. Learning Resources and Events

Admins publish resources and events. Students receive published resources/events; resource progress is persisted. The documentation explicitly notes that event registration exists in the data model but the complete student registration API/UI is not present in the current documented code, so it should be treated as future work.

10. Community, Moderation and Connections

Community posts are analyzed before publication. Spam/fraud phrases, duplicate posts, links, repeated characters and capitalization contribute to risk. Risk under 20 is visible; 20 or above goes to pending review; fraud and spam labels have documented thresholds. Accepted student connections unlock private chat and connected-profile visibility.

11. Real-Time Private Chat

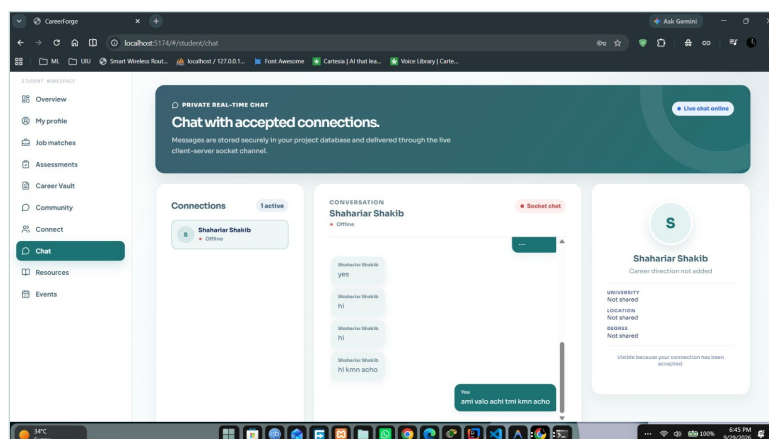


Figure 13. Student private chat interface.

The WebSocket chat flow verifies participants, stores messages, delivers them to active sessions, maintains unread counts, and provides a REST fallback. Only accepted connections can chat.

12. Job Synchronization and Administration

External job synchronization uses source adapters, duplicate detection, status/error tracking and optional scheduling. Remote, configured Bdjobs import, and authorized LinkedIn API/feed integration are documented.

13. Complete Student Journey

Step	Action	Result
1	Register/Login	Account and role validated
2	Complete Profile	Career attributes stored
3	Create Default CV	CV available for matching/applications
4	Open Smart Matches	Recommendations generated
5	Apply	Application + CV snapshot
6	Track Application	Status and match reason
7	Create Learning Path	Topic and levels stored
8	Pass Levels	70%+ unlocks next level
9	Use Resources	Save/complete progress
10	Community/Connect	Participate and build connections
11	Chat	Accepted connections communicate in real time

This follows the documented complete student journey.

14. Screenshots and Demonstrated Results

The screenshots included in this report demonstrate both the Student and Admin workspaces:

Figure	Workspace	Screen
1	Public	Landing page
2	Student	Career Vault
3	Student	Job Matches
4	Student	Assessments and learning roadmap
5	Student	Resources and skill resource finder
6	Admin	Jobs management
7	Admin	Job import/sync progress
8	Admin	Resource management form
9	Admin	Community moderation dashboard
10	Admin	Pending cases and removed posts
11	Admin	Resources overview
12	Admin	Imported YouTube drafts
13	Student	Private real-time chat

15. Future Work

- Complete student-facing event registration.
- Expand learning analytics and assessment types.
- Further improve semantic job-match explanations and ranking coverage.
- Expand structured skill pathways and resource curation.
- Broaden community moderation patterns while retaining admin review.
- Extend connection and messaging capabilities within the accepted-connection privacy boundary.
- Add more authorized external job-source integrations.

16. Conclusion

CareerForge integrates career profiles, CVs, opportunities, applications, learning, resources, community, connections and real-time chat into one student-oriented workspace. Its documented fallback design keeps important workflows usable when external AI services are unavailable. The supplied screenshots demonstrate both Student and Admin workspaces and the major functional areas implemented in the project.